SARVAGYA SINGH

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LinkedIn

EDUCATION

Hong Kong University of Science and Technology *B.Eng. Artificial Intelligence* *Sept 2026–2030*

Delhi Public School, Bokaro (CBSE) *Class XII | SAT: 1460* *2024–2026*
Class X: 96.8% | Class XII Prelim (AISSCE): 96.4%

PROFESSIONAL EXPERIENCE

Full Stack & DevOps Engineer *Education India* *May 2025–Present*

- Architecting edtech platform: course selection, certification, analytics (React, Node.js, PostgreSQL).
- CI/CD with GitHub Actions + Docker; 40% faster deployments/
- Optimised grading microservices (Redis + Celery); latency 12s → <2s.

Backend Engineer *Education India* *Apr 2024–May 2025*

- Built application backend: shortlisting algorithms, interview APIs, status workflows (1,000+ apps).
- REST/GraphQL APIs for quizzes, assignments, auto-grading, certificates; vector DB for recommendations.
- Role-based auth, rate-limiting, audit logging; ensured student data privacy compliance.

Software Engineering Intern *Education India* *Jan–Apr 2024*

- Developed MVP features: form validation, document upload, status updates (React + FastAPI).
- Prototyped quiz engine: question banking, timer logic, MCQ auto-grading.
- Git workflows, agile sprints, production debugging in collaborative environment.

SELECTED HONOURS

- **UNESCO Best Scientific Creativity Award** — Global top prize (3,062 submissions): Developed PDE-constrained PINN loss function & trained photon transport simulator; UNESCO feature.
- **Future Leader Award** — Hong Kong University of Science and Technology: Recognised for leading cross-functional AI deployment in resource-constrained settings; selected from competitive international pool.
- **Global Rank 1** — ACES ISRC (1,000+ teams): Led end-to-end multimodal cardiac AI pipeline & hospital validation protocol; live demo.
- **National Rank 2** — NIT Jamshedpur Conclave: Designed edge-AIoT anomaly detection & deployed telemedicine system; live demo.
- **All-India Rank 15** — WRO Nationals: Built ROS2 navigation stack & real-time YOLO crack detection; featured in 10+ national newspapers.
- **IISER Bhopal Summer Research Fellow** — Top 7% of 3,500 applicants: Mastered rigorous experimental design & reproducibility frameworks for clinical AI validation.

RESEARCH & PROJECTS

Pulsedage — Clinical AI for HCM *Team Member. (3-person team) | pulseage.vercel.app* *2024*

- Architected multimodal CNN ensemble (ResNet-50 + EfficientNet-B4) fusing ECG + MRI for HCM detection, selecting CNN over ViT to meet hospital server compute constraints.
- Engineered DICOM/HL7 preprocessing + class-weighted focal loss to address 1:8 clinical prevalence imbalance; enables reliable detection in low-prevalence screening settings.
- Validated at Bokaro General Hospital (500 beds); 98.5% sensitivity/specificity on 200+ patients; owned model architecture, deployment, & clinician feedback loop.
- Optimised real-time inference API (<200 ms) via INT8 quantization + Vercel edge; enables offline clinical inference on low-resource hospital hardware.

Nandani — Edge-AIoT Livestock Platform *Lead Eng. (3-person team) | gauseva-ai.vercel.app* *2023–2024*

- Designed LSTM anomaly detection on embedded IoT collars (Cortex-M4), pruning model to fit 256KB RAM at <10mW for 24/7 farm operation.
- Implemented offline-first MQTT + SQLite sync architecture; ensures reliable telemetry in <2G rural networks where connectivity is intermittent.
- Deployed low-bandwidth telemedicine system serving 40+ Jharkhand farms; owned edge deployment strategy + IIT (ISM) hardware prototyping coordination.
- Built hardware-aware INT8 quantization pipeline; 40% fewer false positives enables early disease intervention, reducing livestock mortality for smallholder farmers.

PromptBiotics — AI Nutrition Platform *NUS Emergent Ventures (Top 0.4%) | promptbiotics.org* *Jun–Aug 2024*

- Built constitutional AI-aligned RAG system (fine-tuned GPT-4o), using hybrid semantic/BM25 retrieval + cross-encoder reranking to balance relevance vs. latency.
- Developed medical safety validation layer with constraint-based output filtering; 60% hallucination reduction ensures clinical recommendations align with WHO dietary guidelines.
- Beta-launched to 500+ users with active retention tracking; owned prompt engineering, safety evaluation, + VC presentation strategy.
- Integrated user feedback loop for continuous prompt refinement; enables iterative improvement of clinical accuracy based on real-world usage patterns.

Autonomous Inspection Rover *WRO Nationals — AIR 15* *2024*

- Built ROS2 rover with LiDAR + camera fusion for crack detection in hazardous zones, selecting EKF fusion over particle filter for real-time edge GPU performance.
- Implemented A*/DWA path planning + TensorRT-accelerated YOLO; 35 FPS at <50ms latency enables safe autonomous inspection without human entry.
- Owned navigation stack + CV pipeline integration; coordinated 3-person team on sensor calibration, simulation (Gazebo), + field testing.
- Documented full ROS2 architecture; featured in 10+ newspapers for enabling safer infrastructure maintenance in high-risk environments.